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Geospatial and machine learning analyses of cardiovascular disease mortality across the continental United States: Identifying associated variables using Shapley values

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Abstract

Background Cardiovascular diseases (CVDs) remain the leading cause of mortality in the U.S. and exhibit pronounced geographic variation. Although prior studies have documented regional disparities, fewer have combined spatial pattern detection with systematic comparisons of machine learning and deep learning approaches to identify county-level variables associated with CVD mortality at a national scale.

Methods County-level CVD mortality data (2018–2021) were analyzed across the continental U.S. Spatial autocorrelation was examined using Global Moran's I and Getis–Ord Gi* statistics to identify clustering patterns. Separately, predictive modeling was conducted using five machine learning algorithms: linear regression, decision tree, random forest, support vector machine, and extreme gradient boosting, and a deep learning artificial neural network (ANN), drawing on 40 demographic, clinical, socioeconomic, environmental, healthcare, and behavioral variables. Model interpretability was assessed using Shapley Additive Explanations (SHAP).

Results Significant spatial clustering of CVD mortality was observed, with persistent high-mortality hotspots concentrated in the southeastern U.S., consistent with the "Stroke Belt." Among the evaluated models, the ANN achieved the highest predictive performance ($R^2 = 0.89$), followed by XGBoost ($R^2 = 0.82$). SHAP analyses consistently identified hypertension prevalence, population aged 65 years and older, poverty, long-term $PM_{2.5}$ exposure, and rural–urban status as the most influential contributors to CVD mortality predictions.

Conclusions These findings highlight the strong geographic clustering of CVD mortality in the U.S. and demonstrate the value of interpretable predictive modeling for clarifying how multiple, co-occurring population-level factors align with observed spatial disparities. Together, spatial analysis and explainable machine learning provide complementary insights into the distribution of CVD mortality, informing place-based public health assessment.

Keywords Artificial neural networks, Cardiovascular diseases, Machine learning, Shapley, Spatial analysis

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Background

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, accounting for approximately 20.5 million deaths in 2021, corresponding to nearly 32% of global mortality [1, 2]. CVDs encompass a wide range of disorders adversely impacting the heart and blood vessels [3], including coronary heart disease, stroke, heart failure, peripheral arterial disease, and arrhythmia [4]. According to the World Health Organization (WHO), nearly 85% of CVD-related deaths result from heart attacks and strokes [5]. In the U.S. alone, CVDs were responsible for almost 1 million deaths in 2021, equivalent to nearly one death every 33 s [6, 7]. This persistent burden underscores the critical need for comprehensive strategies to identify and control major predictors associated with CVD mortality nationwide.

CVD mortality is influenced by a complex interplay of variables spanning demographic, socioeconomic, behavioral, environmental, and health-related domains. Demographic variables, including age, sex, and race/ethnicity play a central role in shaping cardiovascular risk. Despite overall improvements in population health, persistent racial and ethnic disparities in CVD mortality have been documented, with consistently higher mortality observed among Black populations in the U.S [8, 9]. Socioeconomic disadvantage further exacerbates risks, as lower income [10], unemployment [11], poverty [12], rural-urban disparities [13], and low education attainments [14] are all strongly associated with increased CVD incidence and mortality. Health-related conditions, including obesity, type 2 diabetes mellitus, hypertension, and high cholesterol are also consistently recognized as key drivers of CVD burden. A large-scale cohort study of over 521,000 participants revealed that higher cholesterol intake and whole egg consumption were strongly associated with CVD mortality [15]. Empirical findings also suggested that there are higher risks of CVD death for individuals suffering from type 2 diabetes mellitus and obesity [16].

Beyond clinical predictors, growing evidence highlights the long-term associations between behavioral and environmental exposures and CVD outcomes. Behavioral factors such as physical inactivity and smoking have been shown to independently and synergistically increase CVD mortality risk, particularly among older adults with hypertension [17]. Environmental exposures, especially air pollution, have also emerged as critical determinants of cardiovascular health. Prior research has demonstrated that both short-term and long-term exposure to ambient air pollutants are significantly associated with increased risks of ischemic heart disease and cardiovascular mortality [18].

Given the multifactorial nature of CVD mortality and their interdependencies, accurately quantifying their relative contributions is challenging. Traditional epidemiological approaches provide valuable insights but often struggle to capture nonlinearities and complex interactions. Advances in artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), offer robust alternatives [19]. These data-driven modeling approaches can uncover hidden patterns among predictors and outcomes, leading to better accuracy and more reliable risk prediction [20]. Applications of ML and DL in public health have expanded rapidly since the early 2000s, encompassing infectious disease modeling [21], assessment of healthcare accessibility disparities [22], development of cancer treatment strategies [23], and identification of key risk factors [24].

ML and DL techniques have demonstrated promising performance in CVD research for prediction and predictor identification. A recent study conducted on a 10-year CVD incidence data comparing the predictive power of ML classifiers with the traditional risk prediction tool of HellenicSCORE (a calibration of the ESC SCORE) found that ML approaches more effectively captured complex interrelationships among predictors [25]. Similarly, findings from a 13-year follow-up study on 6,459 participants [26] reported that employing advanced ML modeling approach, such as support vector machines (SVM) outperformed the current CVD risk calculator of the American College of Cardiology/American Heart Association. In another effort to forecast CVD mortality in Pakistan, Qureshi et al. [27] showed that an artificial neural network autoregressive (ANNAR) model outperformed conventional time-series approaches in forecasting CVD mortality by better capturing nonlinear patterns and disease dynamics.

Geospatial analysis further enhances CVD research by explicitly accounting for geographic variation and spatial dependence in disease outcomes [28]. Common spatial autocorrelation methods such as Moran's I and Getis–Ord Gi* facilitate the identification of geographic clusters, detection of high-risk regions, and examination of contextual influences on disease outcomes [29]. A recent global analysis of CVD mortality and its associated risk factors also highlighted the importance of geospatial modeling applications [30]. Their geospatial analysis showed that CVD mortality has followed an increasing trend in both developed and developing countries, with socioeconomic and behavioral factors positively associated with higher mortality rates.

Despite these methodological advances, much of the existing literature on CVD mortality has relied on a limited set of covariates and a narrow range of modeling approaches, often without systematic comparison across algorithms. To address these gaps, we incorporated a comprehensive set of 40 potential variables categorized into six different groups, including demographic, clinical,

socioeconomic, environmental, healthcare delivery and behavioral predictors. We applied and compared predictive performance of six modeling approaches, including linear regression (LR), decision tree (DT), SVM, random forest (RF), extreme gradient boosting (XGBoost), and artificial neural network (ANN). Then, we compared their predictive performance using root mean square error (RMSE), mean absolute error (MAE), and R-squared (R^2). To enhance interpretability, we employed SHapley Additive exPlanations (SHAP) to identify and rank the most influential predictors of county-level CVD mortality across the continental U.S.

Methodology

Data collection

County-level CVD mortality data were obtained from Centers for Disease Control and Prevention (CDC) Atlas of Heart Disease and Stroke [31] from the beginning of January 2018 until the end of December 2021. This dataset contains annual mortality rates (per 100,000 population) for each county across the continental U.S. Mortality data were aggregated into four consecutive one-year intervals: January–December 2018, January–December 2019, January–December 2020, and January–December 2021.

In addition to CVD mortality data, a set of 40 county-level variables was extracted from the CDC and categorized into six groups: demographic, clinical, socioeconomic, environmental, healthcare delivery, and behavioral predictors, as the explanatory variables. Detailed information is presented in Table 1. All mortality and explanatory variable datasets were spatially joined to the U.S. county boundary shapefile obtained from the U.S. Census Bureau website.

Spatial analysis

Global Moran's I, a spatial autocorrelation technique, was applied within the ArcGIS Pro 3.6 (Esri, Redlands, CA) environment to examine the spatial distribution of CVD mortality rates across the continental U.S. Moran's I values range from − 1 to + 1, representing clustered (> 0), dispersed (< 0), or random (= 0) distributions [32]. In addition, local spatial autocorrelation Getis–Ord Gi* was employed to assess the statistical significance of spatial clusters of high and low mortality rates. The Gi* statistic identifies concentrated clusters of CVD mortality rates through z-scores [33, 34]. A county with a statistically significant positive Gi* indicates a hotspot, typically surrounded by neighboring counties with similarly high CVD mortality. In contrast, a statistically significant negative z-score indicates a cold spot, where counties with lower CVD mortality are clustered together. Larger absolute z-scores indicate stronger degrees of spatial clustering.

Data preprocessing

Missing values in explanatory variables were handled using mean imputation, whereby each missing value was replaced with the arithmetic mean of the observed values for the corresponding variable. This approach preserved the full analytic sample while minimizing the potential risk of bias in parameter estimates, as missingness was minimal across variables. Given the inclusion of a large set of potentially correlated predictors (n = 40), variance inflation factor (VIF) and Pearson correlation analyses were conducted to assess and reduce multicollinearity. By doing this, we can improve the generalizability of the models by reducing the risk of overfitting. Variables with VIF values exceeding a threshold of 7.5 were considered highly collinear and excluded, with the most informative and least correlated variables retained as model inputs [24]. Accordingly, six predictors were excluded due to high multicollinearity (all with VIF values > 7.5), and the remaining 34 predictors were retained as the final set of model inputs for subsequent models. The excluded variables and their corresponding VIF values are reported in Additional File 1 (Table S1). All explanatory variables were subsequently normalized using Min–Max scaling to ensure a common numerical range, thereby facilitating efficient model training and preventing disproportionate influence of variables with larger scales. This normalization step was particularly important for ANN, as it supports stable convergence during training and reduces the risk of gradient saturation [35]. To identify anomalous observations in the input data, a density-based spatial clustering algorithm (DBSCAN) was applied for grouping data records that are closely packed (core groups), while identifying records in low-density regions, known as outliers or noise points. DBSCAN helps enhance interpretability by flagging potential outliers prior to model fitting [36]. In our analysis, DBSCAN did not label any counties as outliers or noise points (0 outliers among 3,109 counties; 0%). Therefore, no counties were excluded and the full sample was retained for subsequent modeling.

Analytical framework

The primary objective of our study is to model CVD mortality rates (target) based on a comprehensive set of explanatory variables (features) by implementing a combination of ML and DL algorithms. For this purpose, we randomly split the data into training (80%) and testing (20%) subsets to build and evaluate out-of-sample performance, respectively. We then employed a comparative modeling framework consisting of five ML techniques and a DL model based on ANN topology (Fig. 1). Brief descriptions of each model are outlined below:

Table 1 County-level explanatory variables included in the analysis, categorized by demographic, clinical, socioeconomic, environmental, healthcare delivery, and behavioral domains

Category	Variable	Description
Demographic (10 variables)	Population ages 65 and above	Population Ages 65 and above, (% of total population), 2018–2022 (5-year)
	Population ages 65 and above, male	Male population Ages 65 and above (% of male population), 2018–2022 (5-year)
	Population ages 65 and above, female	Female population Ages 65 and above (% of female population), 2018–2022 (5-year)
	Asian (non-Hispanic), all ages (%)	Asian, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	American Indian/Alaska Native (non-Hispanic), all ages (%)	American Indian/Alaska Native, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	Black (non-Hispanic), all ages (%)	Black or African American, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	White (non-Hispanic), all ages (%)	White, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	Some Other Race (non-Hispanic), all ages (%)	Some Other Race, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	Two or More Races (non-Hispanic), all ages (%)	Two or More Races, non-Hispanic Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
	Hispanic, all ages (%)	Hispanic/Latino Population as a Percentage of Total Population, All Ages, 2018–2022 (5-year)
Clinical (4 variables)	High Cholesterol (%)	High Cholesterol Among Adults Screened in Past 5 Years Ages 18+, 2021
	Diabetes (%)	Diagnosed Diabetes, Age-Adjusted Percentage, 20+, 2021
	Obesity (%)	Obesity, Age-Adjusted Percentage, 20+, 2021
	High Blood Pressure (%)	Hypertension Rate per 100,000, All Races/Ethnicities, Both Sexes, Ages 35+, 2019–2021
Socioeconomic (10 variables)	Education – Less than High School (%)	Percentage without High School Diploma, Ages 25+, 2018–2022 (5-year)
	Education – Less than College (%)	Percentage without 4 + Years College, Ages 25+, 2018–2022 (5-year)
	Median Household Income ($)	Median Household Income, 2021
	Income Inequality	Income Inequality (Gini Index), 2018–2022 (5-year)
	Poverty (%)	Percentage Living in Poverty, All Ages, 2021
	Severe Housing Cost Burden (%)	Households That Spend 50% or More of Their Household Income on Housing (%), 2017–2021 (5 year)
	Unemployment Rate (%)	Unemployment Rate, Ages 16+, 2022
	Rural-Urban Status	Urban-Rural Status, 2021
	Health Insurance Status (%)	Percentage without Health Insurance, Under Age 65, 2021
	Median Home Value ($)	Median Home Value, 2018–2022 (5-year)
Environmental (2 variables)	Air Quality (Annual $PM_{2.5}$ Level)	Annual Average Ambient Concentrations of $PM_{2.5}$, 2019
	Park Access (%)	Percentage of Population Living Within Half a Mile of a Park, 2020
Healthcare delivery (12 variables)	Cholesterol Screening (%)	Cholesterol Screening Among Adults Ages 18+, 2021
	Blood Pressure Medication Use (self-report) (%)	Blood Pressure Medication Use Among Adults with High Blood Pressure Ages 18+, 2021
	Nonadherence – Diuretic only (%)	Diuretic Nonadherence Percentage, Medicare Part D Beneficiaries, Ages 65+, 2019
	Nonadherence – Renin-Angiotensin System only (%)	Renin-Angiotensin System Antagonist Nonadherence Percentage, Medicare Part D Beneficiaries, Ages 65+, 2019
	Cardiac Rehabilitation Use Eligibility Rate	Cardiac Rehabilitation Eligibility Rate per 1,000 Medicare Fee-for-Service Beneficiaries, 2018–2019
	Cardiac Rehabilitation Use Completion Rate Among Initiators (%)	Cardiac Rehabilitation Completion Rate Among Initiating Fee-for-Service Medicare Beneficiaries, 2018–2019
	Primary Care Physicians	Population per Primary Care Physician (in thousands), 2021
	Cardiovascular Disease Physicians	Population per Cardiovascular Disease Physician (in thousands), 2021
	Hospitals (#)	Number of Hospitals per 100,000 population, 2021
	Pharmacies and Drug Stores (#)	Number of Pharmacies and Drug Stores per 100,000 population, 2021
	Neurologists	Population per Neurologist (in thousands), 2021
	Neurosurgeons	Population per Neurosurgeon (in thousands), 2021

Table 1 (continued)

Category	Variable	Description
Behavioral (2 variables)	Physical Inactivity (%)	Leisure-Time Physical Inactivity, Age Adjusted Percentage, 20+, 2021
	Smoking (%)	Current Smoker Status Among Adults Ages 18+, 2021

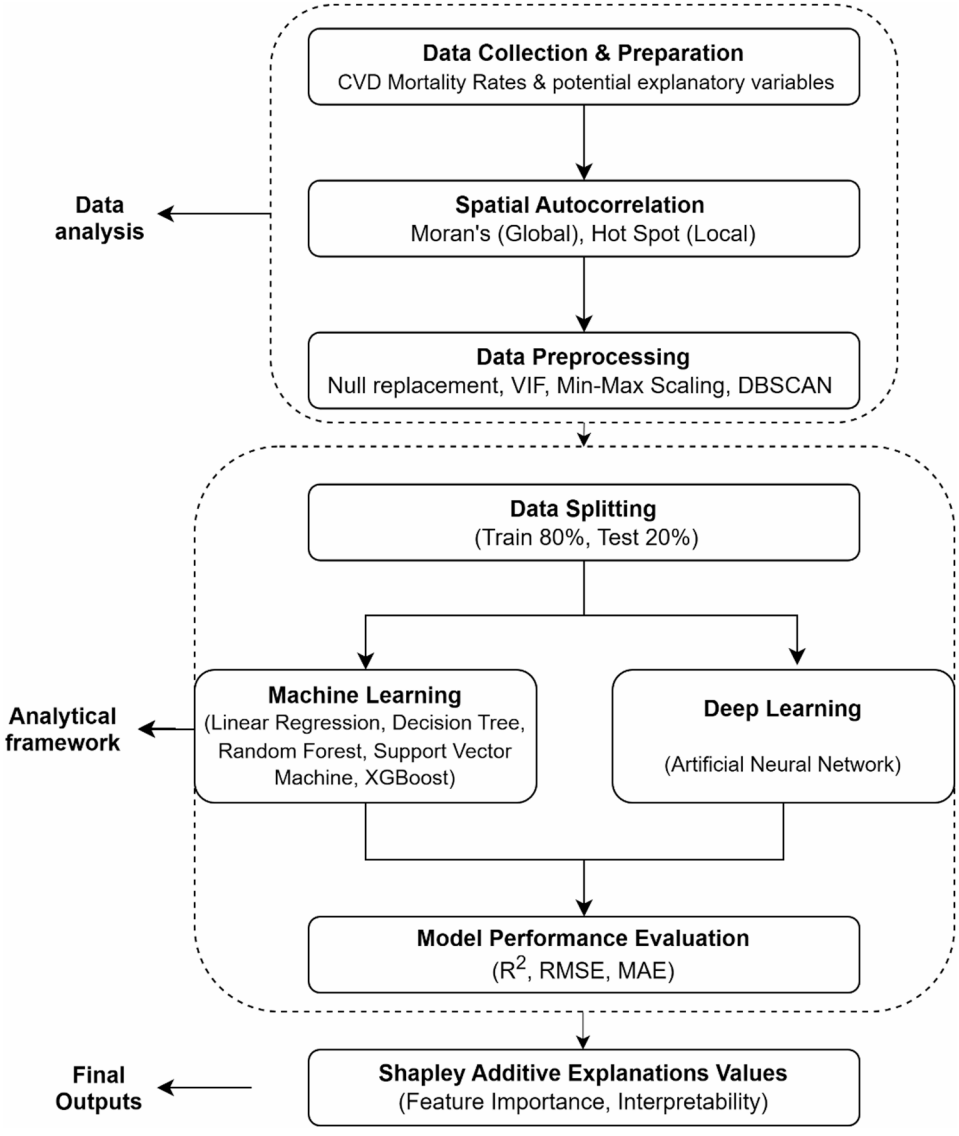


Fig. 1 Methodological flowchart of the study

Linear Regression

As a baseline model, a simple parametric linear regression (LR) was applied to model CVD mortality rates as a linear combination of predictors. Coefficients were estimated under assumptions of linear relationships and normally distributed errors. Each coefficient reflects the adjusted association of its corresponding determinant with the outcome [37].

Decision Tree

A DT regressor was employed as a non-parametric supervised learning algorithm to capture complex nonlinear patterns between predictors and CVD mortality rates. This approach recursively partitions the dataset based on predictors values, constructing a tree in which each internal node represents a decision rule and each leaf corresponds to a predicted CVD mortality value. The hierarchical structure facilitates interpretability and identification of influential variables [38].

Random Forest

Building upon the DT approach, RF, an ensemble learning algorithm, constructs multiple DTs during training, each on a random subset of the data and aggregates their predictions. This ensemble approach enhances predictive accuracy and mitigates overfitting by averaging across trees, and provides variable-importance measures that aid in identifying crucial factors for CVD mortality [39].

Support Vector Machine

SVM, as a supervised non-linear learning technique, identifies an optimal decision boundary in the transformed feature space to perform regression tasks. By utilizing kernel functions, SVM can effectively handle intricate relationships between predictors and outcome [40].

eXtreme Gradient Boosting

XGBoost, an advanced gradient-boosting algorithm, builds additive trees sequentially. Its objective function includes an explicit regularization term to penalize complexity, thereby improving generalization and reducing overfitting [41]. By iteratively adding trees to correct residual errors, XGBoost captures higher-order interactions among CVD-associated variables and enhances predictive accuracy.

Artificial Neural Network

In addition to ML models, we implemented a DL approach using an ANN to model the relationships between the set of predictors and CVD mortality. ANNs are computational models inspired by biological neural systems, consisting of interconnected processing units, or neurons, arranged in layers [42]. These models are particularly suitable for capturing complex, nonlinear relationships in high-dimensional data that may not be adequately represented by conventional statistical methods. A deep feedforward ANN architecture was implemented using TensorFlow, consisting of an input layer, six hidden layers, and a single output layer (Fig. 2).

The dense hidden layers consisted of 256, 128, 64, 32, 16, and 4 neurons respectively. All layers were fully connected and employed Rectified Linear Unit (ReLU) activation function. ReLU was selected for its non-saturating nature, which mitigates the vanishing gradients and supports efficient gradient propagation in deep architectures [43]. Each neuron computes a weighted linear combination of inputs plus a bias term:

$$Z_i = \sum_{j=1}^{n} w_{ij} x_j + b_i \quad (1)$$

Where x_j are the explanatory variables, w_{ij} is the learned weights, and b_i is the bias term. Then, the ReLU activation function is applied:

$$ReLU\ (z) = \max(0,\ z) \quad (2)$$

To improve training stability and reduce internal covariate shift, batch normalization was applied after each hidden layer [44]. The output layer consisted of a single neuron with a linear activation function, corresponding

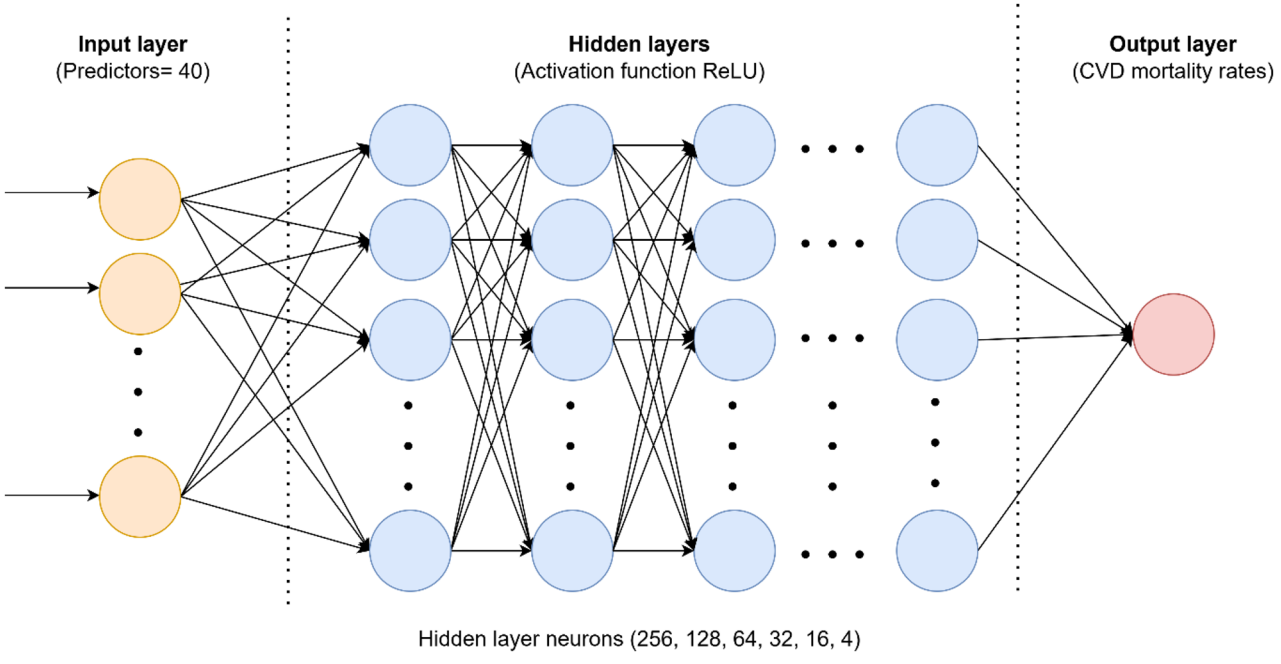


Fig. 2 ANN topology used in this study

to the continuous outcome variable (i.e., CVD mortality rate). The network was compiled using the RMSprop optimizer with a learning rate of η =0.002, chosen for its robustness in handling non-stationary optimization landscapes typical in DL applications [45]. We monitored mean absolute error (MAE), RMSE, and R^2 to evaluate model performance on both training and test datasets. The network was trained for 10 epochs using mini-batch gradient descent, with weights updated according to Eq. 3:

$$w_{ij}^{(n+1)} = w_{ij}^{(n)} - \eta \frac{\partial L^{(n)}}{\partial w_{ij}{}^{(n)}} \quad (3)$$

where $w_{ij}^{(n)}$ is the weight from neuron j to neuron i at the nth iteration. η is the learning rate and $\frac{\partial L^{(n)}}{\partial w_{ij}{}^{(n)}}$ is the gradient of the loss function with respect to corresponding weight. The dataset was split into training (80%) and testing (20%) subsets, with 15% of the training data reserved for validation. Early stopping criteria was applied to avoid overfitting. To assess training stability and sensitivity to the number of epochs, we also trained the ANN for 20, 50, and 100 epochs and observed no meaningful changes in validation and test performance, indicating stable model behavior and no evidence of overfitting. Training dynamics were monitored using TensorBoard to visualize loss trajectories and performance metrics in real time.

Model performance evaluation

We assessed each model's predictive accuracy using three metrics: R^2, MAE, and RMSE. R^2 quantifies the proportion of variance in observed CVD mortality rates explained by the model. MAE calculates the average absolute differences between predicted and actual values, and RMSE measures the square root of the average squared residuals. The metrics are defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y_i})^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2} \quad (5)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|(y_i - \widehat{y_i})| \quad (6)$$

where $\widehat{y_i}$ are the predicted values, y_i are the actual values, and $\bar{y}$ is the mean value of the observed outcomes.

Moreover, spatial autocorrelation of residuals from the models was evaluated to assess whether any remaining spatial dependence persisted after model fitting. Global Moran's I was computed for the model residuals. If Moran's I is non-significant, residuals exhibit no meaningful spatial clustering, suggesting that predictive performance is unlikely to be inflated by unmodeled spatial dependence.

SHAP-based model interpretability

To enhance interpretability and quantify individual predictor contributions, we employed SHAP, a theoretically grounded framework based on cooperative game theory. SHAP treats each predictor as a "player" in a cooperative game where the "payout" is the model's prediction minus the average prediction. Each feature's SHAP value represents its average marginal contribution across all possible subsets of features. This ensures three properties, including local accuracy (sum of contributions equals the prediction deviation), consistency (if a model's dependence on a feature increases, its attributed value does not decrease), and missingness (feature not present receives zero attribution). The SHAP values (Φ_i) are calculated as follow:

$$\Phi_i = \sum_{S \subseteq N \setminus \{i\}} \left(\frac{|S|!\,(|N| - |S| - 1)!}{|N|!}\right) \left[V_{S\cup\{i\}}(S \cup \{i\}) - V_S(S)\right] \quad (7)$$

where N is the total number of features, the value function V_S denotes the model's expected prediction using the variables in coalition S. This formulation measures feature i's average marginal contribution across all coalitions.

For model interpretability, SHAP values were computed for the ANN models using the permutation-based SHAP explainer, which provides a model-agnostic and robust approximation of Shapley values for complex, nonlinear models [46]. For each year (2018–2021), SHAP analyses were conducted separately using the corresponding year-specific trained ANN. The background dataset for SHAP was defined as the full training set for each respective year, representing the empirical distribution of county-level characteristics learned by the model. This background was used to estimate the expected model output and to compute both global and local SHAP values. Consistent explainer settings were applied across all years to ensure comparability of SHAP-based feature importance and effect patterns over time.

We summarized SHAP results using global feature importance bar plots (mean absolute SHAP values per feature), beeswarm plots illustrating magnitude and direction of effects, and force plots providing local explanations of individual predictions. We also generated SHAP dependence plots for selected top predictors to visualize how predictor values relate to their SHAP contributions and to highlight potential nonlinear relationships. The SHAP analyses enabled ranking of influential

variables, visualization of whether a feature increases or decreases predicted CVD mortality, and interpretation of localized feature contributions at the county level.

Results

Spatial analysis findings

Global Moran's I revealed statistically significant positive spatial autocorrelation in CVD mortality rates (Moran's I = 0.45, $p < 0.01$), indicating that CVD mortality rates are not randomly distributed but spatially clustered. Counties with elevated CVD mortality tended to be surrounded by neighboring counties with similarly high rates, confirming spatial dependence in CVD outcomes. As shown in Fig. 3A, the southeastern region of the U.S., often referred to as the "Stroke Belt", demonstrated persistently elevated mortality rates across both years, with limited differences over time. Higher mortality rates are observed in Alabama, Arkansas, North Carolina, South Carolina, Georgia, Louisiana, Mississippi, and Tennessee. Local hotspot analysis (Fig. 3B) further revealed that high-mortality clusters were concentrated primarily in the Southeast, including large portions of the Appalachian region and Deep South. The intensity and spatial extent of these hotspots expanded between 2018 and 2021, with a notable increase in the number of counties falling under the 99% confidence interval. Notably, clusters of high mortality rates were apparent in counties across Mississippi, Alabama, Kentucky, Arkansas, and parts of Louisiana and Tennessee.

In contrast, significant cold spots were observed in counties located in the Midwest, the Mountain West, and Northeast, suggesting more favorable CVD outcomes in those regions. Counties across the Northeast (New York, Pennsylvania, Vermont, and New Hampshire) and Mountain West (Utah, Montana, and Idaho) regions consistently reported lower CVD mortality rates. Additional cold spots were observed in the upper Midwest, particularly in Minnesota, Michigan, and Wisconsin (Fig. 3B).

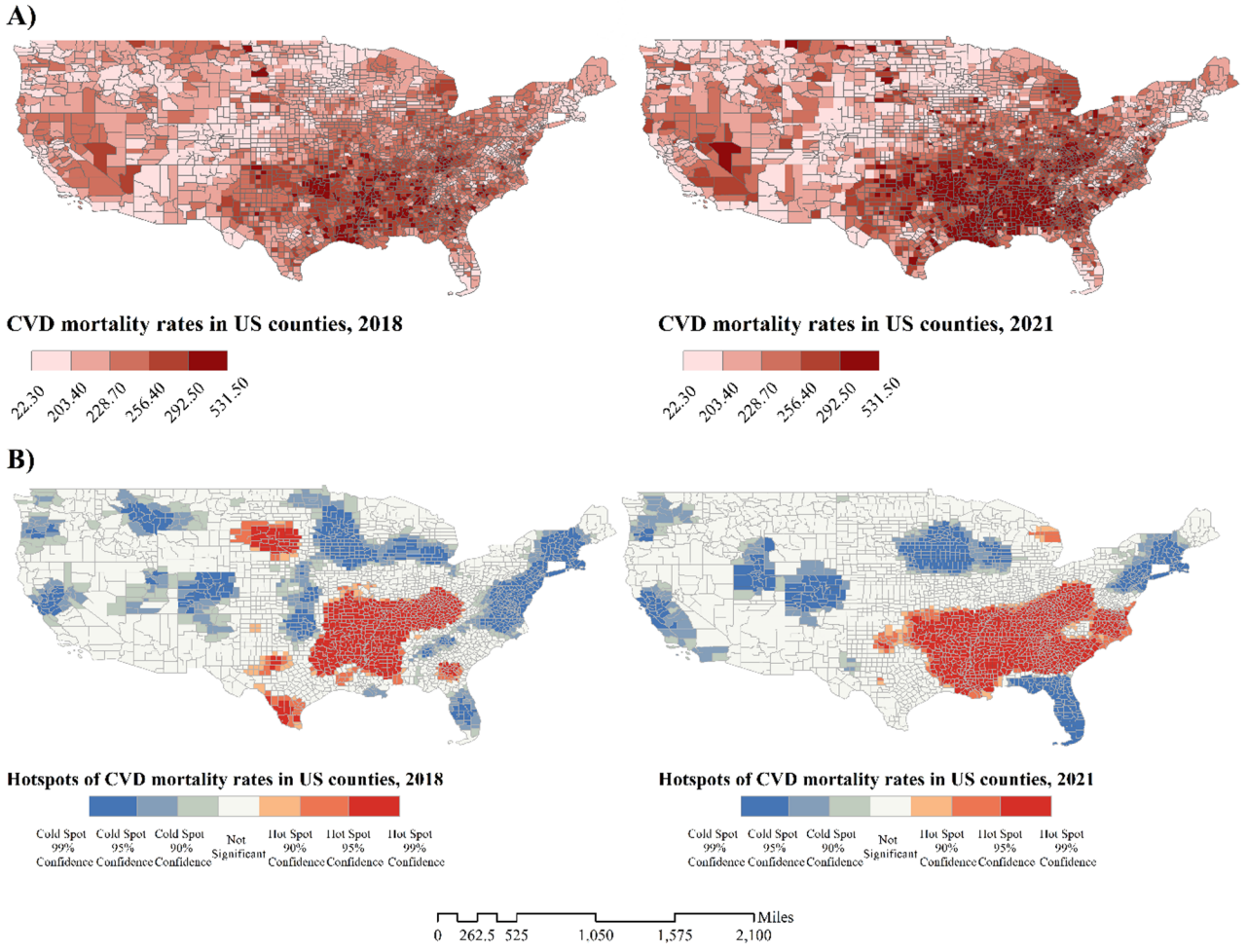


Fig. 3 Geospatial distribution of (**A**) CVD mortality rates in U.S. counties in 2018 (left) and 2021 (right), and (**B**) spatial clusters of CVD mortality rates in 2018 (left) and 2021 (right)

Table 2 Comparison of goodness of fit for LR, DT, RF, SVM, XGBoost, and ANN models

Criterion	LR	DT	RF	SVM	XGBoost	ANN
R^2	0.58	0.70	0.72	0.79	0.82	0.89
RMSE	0.06	0.05	0.05	0.04	0.04	0.03
MAE	0.04	0.04	0.04	0.03	0.03	0.02

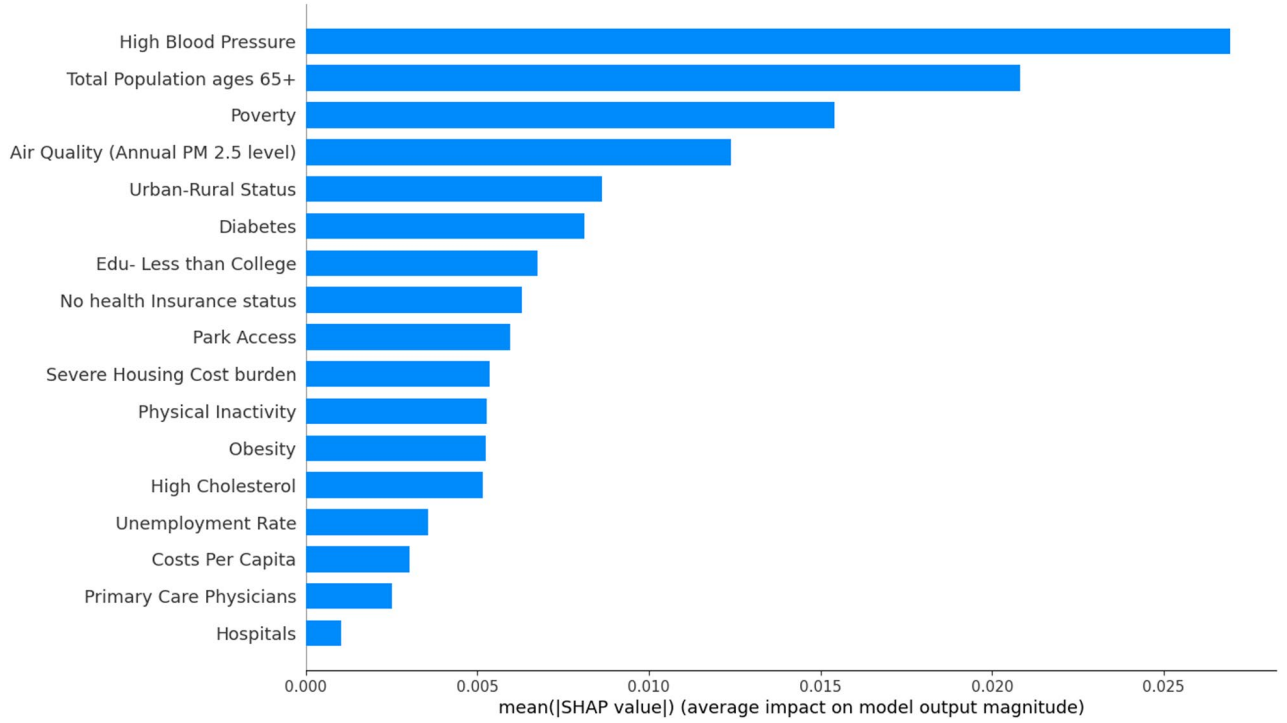


Fig. 4 Ranking of the most influential county-level CVD mortality predictors based on mean SHAP values for 2021

Model performance

Among the evaluated models, ANN demonstrated superior performance. ANN achieved the highest R^2 (0.89), which indicates that it explained approximately 89% of the variance in CVD mortality rates. It also achieved the lowest error metrics (RMSE = 0.03 and MAE = 0.02), reflecting superior model precision. XGBoost, with an R^2 of 0.82, was the second-best performing model, followed by SVM with an R^2 of 79%. The ensemble RF and DT models showed moderate performance, with R^2 values of 0.72 and 0.70, respectively. LR, used as a baseline model, exhibited the lowest explanatory power (R^2 = 0.58) (Table 2). Moreover, Global Moran's I computed on model residuals was non-significant for ANN (Moran's I = 0.006, $p = 0.23$) and XGBoost (Moran's I = -0.002, $p = 0.73$), indicating a spatially random error pattern and suggesting that predictive performance is unlikely to be inflated by residual spatial dependence.

SHAP-based interpretation

SHAP values were computed based on the ANN approach, given its superior predictive performance. Global feature importance results based on the mean absolute SHAP values identified high blood pressure, population aged 65 years and older, and poverty as the top three predictors, followed by ambient air pollution (annual $PM_{2.5}$ levels) and rural–urban status (Fig. 4).

The SHAP beeswarm plot (Fig. 5) illustrates both the direction and magnitude of each predictor's effect across counties. Each point represents a county with position along the x-axis indicating the direction and strength of the feature's contribution to predicted mortality, and color reflecting the feature value from blue (low feature value) to red (high). High blood pressure, population aged 65 years and older, and poverty show pronounced spreads toward positive SHAP values, particularly at higher feature levels, indicating their strong associations in increasing predicted CVD mortality. Conversely, variables such as number of hospitals and physicians show narrower distributions near zero, suggesting more limited predictive influence. SHAP-based feature importance rankings and beeswarm plots for 2018, 2019, and 2020 are provided in Additional File 2 (Figures S1-S6). In addition, SHAP dependence plots for the top four predictors (high blood pressure, population aged 65 years and older, poverty, and annual $PM_{2.5}$) further illustrate

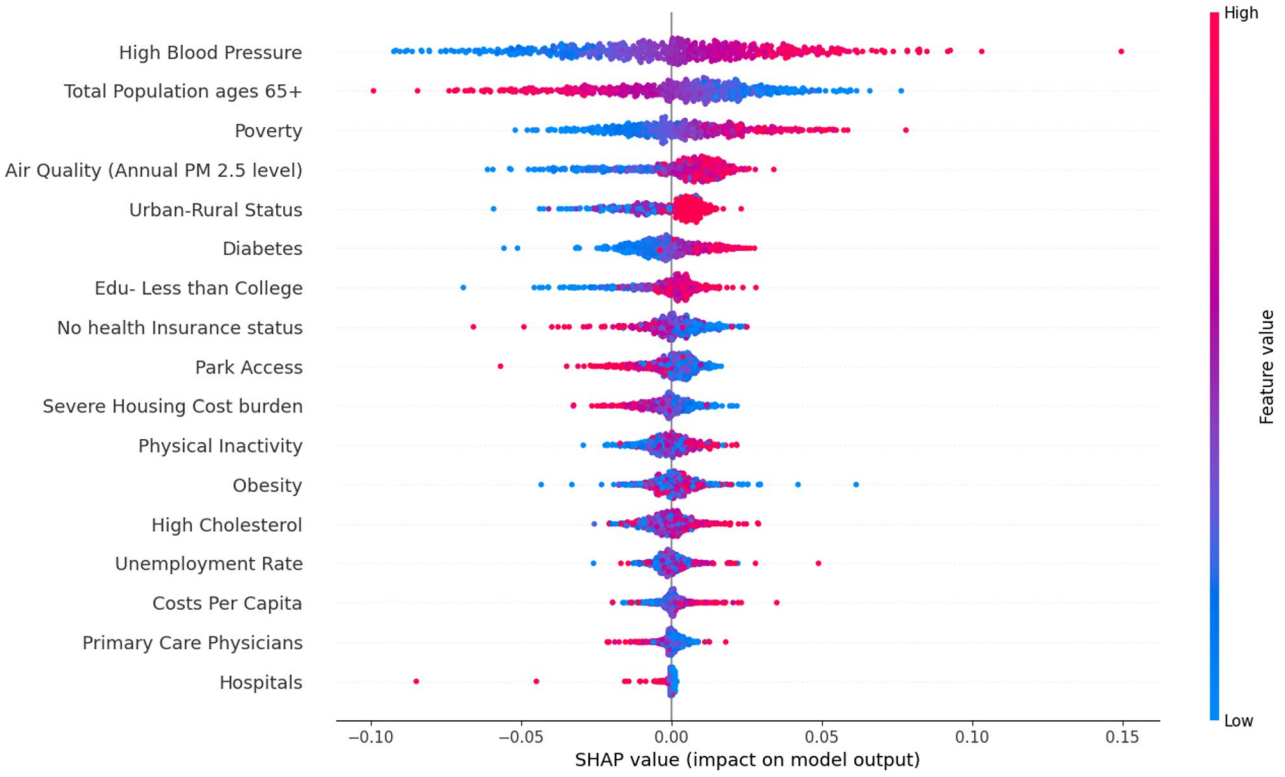


Fig. 5 SHAP beeswarm plot showing the direction and magnitude of effects for the most influential county-level CVD mortality predictors, 2021



Fig. 6 Local SHAP force plot showing feature contributions to predicted CVD mortality in a representative high-risk county, 2021

that higher predictor values are generally associated with increasingly positive SHAP contributions to the ANN-predicted CVD mortality rate (Additional File 3; Figures S1-S5).

Local interpretability is illustrated through a SHAP force plot for a representative high-risk county in 2021 (Fig. 6). The base value (approximately 0.43) indicates the average model-predicted mortality, while the final value (0.59) reflects the county's predicted CVD mortality after accounting for individual feature contributions. Red bars denote features that increase the predicted mortality (e.g., high blood pressure, poverty, older adults' population, and limited park access), while blue bars represent features that reduce the prediction (e.g., lower diabetes prevalence or better insurance coverage). The width of each bar reflects the magnitude of that feature's contribution. This localized explanation demonstrates how multiple features jointly contribute to the model's predicted CVD mortality for a specific county, enhancing interpretability and clarifying key contributors of the prediction.

Discussion

CVDs remain the leading cause of mortality in the U.S., accounting for nearly one-third of all deaths nationwide. This burden reflects the convergence of interconnected demographic, clinical, socioeconomic, environmental, healthcare delivery, and behavioral predictors that vary markedly across geographic contexts. Motivated by this complexity, we employed a comprehensive analytical framework with a broad range of predictors. These predictors were selected based on prior epidemiological evidence and domain knowledge, many of which, especially environmental and behavioral variables, have been underexplored at this geographic scale. To meet the need for an integrated geographic approach, we applied geospatial clustering, five ML models, a DL architecture (ANN), and SHAP-based interpretability tools to model and explain CVD mortality across the continental U.S. We found significant geographic clustering of CVD mortality, particularly in the Southeast, reaffirming the persistence of the "Stroke Belt" [47], highlighting the importance of regional and structural determinants.

Among the six algorithms evaluated, the ANN topology demonstrated the highest predictive accuracy, underscoring the value of DL in capturing complex nonlinear associations at the population scale. This design may improve public health policymakers understanding of spatial disparities and generates interpretable, county-specific insights into the most important predictors.

The predictive superiority of the ANN aligns with a growing body of research demonstrating the advantages of DL in modeling CVD outcomes [48]. Rehman et al. applied ANN techniques to coronary heart disease and reported high prediction accuracy (typically exceeding 95%), surpassing traditional ML methods such as SVM, RF, KNN, and DT [49]. Qureshi et al. showed that an autoregressive ANN model (ANNAR) outperformed conventional time-series forecasting methods for CVD mortality in Pakistan [27]. At the clinical level, deep neural survival models have outperformed Cox proportional hazard models in predicting CVD events, demonstrating improved discrimination and explained variance [50]. While most prior studies have been mostly focused on individual-level or clinical aspects of CVDs, our results extend this evidence to population-level, highlighting the value of ANN in capturing nonlinear and spatial dependence across the continental U.S.

Our SHAP-based findings further align with epidemiological evidence. National mortality trend analyses (2010–2019) have indicated that hypertension-related CVD death rates increased in more than 85% of U.S. counties, particularly among older adults in the Southeast, consistent with our identification of hypertension as a leading variable [51]. CDC data shows that poverty independently magnifies CVD risk, especially in combination with hypertension across racial and ethnic groups [52]. Moreover, long-term $PM_{2.5}$ exposure elevates CVD and stroke mortality, disproportionately affecting older adults, socially disadvantaged communities, and predominantly Black counties [53–55]. Analyses of over 2,000 counties demonstrate that the combined burden of air pollution and poverty contributes to greater life expectancy loss in disadvantaged regions, supporting the spatial patterns observed in our study [56]. The spatial concurrence of these factors in our analysis reinforces the importance of structural and environmental contexts, contributing to CVD mortality rates.

The identification of rural-urban status as a major contributor reflects well-documented rural–urban disparities in CVD outcomes. By 2019, rural death rates were approximately 21% higher than urban rates (189.1 vs. 156.3 per 100,000) [57]. From 2010 to 2022, mortality declined in urban areas but increased among rural adults aged 25–64 [58]. By 2022, rural residents were dying from CVD at 1.5 times the rate of their urban counterparts [59]. Rural adults also exhibit higher prevalence of hypertension, obesity, and diabetes, that might have contributed to earlier and more severe disease onset, consistent with our findings [60]. The National Institute of Health states that socioeconomic and educational disadvantages in rural communities significantly contribute to poorer CVD outcomes [61]. However, not all evidence aligns fully with our results. For example, high cholesterol did not rank among the top contributors after accounting for socioeconomic and spatial factors. This inconsistency likely reflects differences in analytic scale, where individual-level studies capture metabolic risks more directly, whereas county-level models emphasize structural and contextual determinants such as poverty and rurality.

In this study, SHAP played a central role in enhancing the interpretability of our modeling framework. Unlike traditional feature importance metrics that provide only global rankings [62], SHAP decomposes each prediction into additive contributions from all predictors, preserving both directionality and magnitude. This enabled us to capture complex, nonlinear interactions such as the compounded influence of age 65+, hypertension, poverty, $PM_{2.5}$ exposure, and rurality, and to quantify their collective contributions across all counties. Importantly, the major predictors identified through SHAP were not only statistically significant but also aligned with established epidemiological pathways, reinforcing the model's validity. This integration moves beyond "black box" modeling by revealing why certain counties exhibit elevated risk, enabling the translation of complex predictive outputs into insights that are interpretable and actionable for public health stakeholders. From a policy perspective, this integrated framework highlights counties where multiple vulnerabilities converge, supporting more targeted and equitable allocation of public health resources, including interventions focused on hypertension management, healthcare access, and environmental risk reduction in structurally disadvantaged regions.

Despite its strengths, this study has several limitations. First, reliance on county-level data, while appropriate for national-scale analysis, obscures substantial within-county heterogeneity. Future research should incorporate neighborhood-level or census-tract analyses to more effectively capture local disparities. Second, reliance on annual aggregates may mask short-term or seasonal dynamics in CVD mortality and its contributing variables, suggesting value in higher-frequency temporal analyses. Third, several predictors were available only as single-year estimates or multi-year summaries, whereas CVD mortality was analyzed annually from 2018 to 2021. This temporal mismatch may attenuate associations for predictors that vary more rapidly over time and may blunt the model's ability to capture short-term trend changes. As a robustness check, we conducted a sensitivity analysis restricted to 2021 mortality, which

yielded consistent results in model performance and an unchanged ranking of influential predictors. However, future work would benefit from annually harmonized predictors and explicitly lagged time-series predictors to better evaluate short-term dynamics. Fourth, mean imputation and the exclusion of multicollinear variables, although necessary for model stability, may limit the robustness of the findings and could influence SHAP-based rankings. More advanced approaches, such as multiple imputation or recursive feature elimination, could further strengthen robustness. Finally, as demographic profiles, environmental exposures, and healthcare access patterns evolve, periodic model retraining with updated datasets will be essential to maintain predictive relevance and ensure timely, evidence-based public health interventions.

Conclusion

In conclusion, our findings indicate that CVD mortality in the U.S. is characterized by the spatial co-occurrence of demographic aging, chronic disease burden, socioeconomic disadvantage, environmental exposure, and rural context, rather than by isolated factors. While DL demonstrated strong predictive performance, the primary contribution of this study is showing how interpretable modeling can translate complex population-level patterns into clear, geographically grounded insights. The consistency between SHAP-identified contributors and prior epidemiologic evidence supports the plausibility of these patterns and suggests they are not artifacts of model specification. Overall, the results highlight the value of place-based perspectives that consider multiple, overlapping vulnerabilities simultaneously. As population characteristics and exposures continue to change, approaches that pair strong predictive performance with clear interpretability can help guide more targeted and equitable efforts to reduce cardiovascular mortality.

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
ANNAR	Artificial Neural Network Auto-Regressive
CDC	Centers for Disease Control and Prevention
CVD	Cardiovascular Disease
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DL	Deep Learning
DT	Decision Tree
ESC	European Society of Cardiology
GIS	Geographic Information Systems
KNN	K-Nearest Neighbors
LR	Linear Regression
MAE	Mean Absolute Error
ML	Machine Learning
MLP	Multilayer Perceptron
$PM_{2.5}$	Particulate Matter with diameter ≤ 2.5 micrometers
RF	Random Forest
RMSE	Root Mean Square Error
ReLU	Rectified Linear Unit
SHAP	Shapley Additive Explanations
SVM	Support Vector Machine
U.S.	United States
WHO	World Health Organization
XGBoost	Extreme Gradient Boosting

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Authors' contributions

NK, SD, and MT contributed to the study design and conceptualization. NK, MT, and AM performed data preprocessing and contributed to model development, including additional analyses conducted during the revision stage to address reviewers' comments. NK drafted the initial version of the manuscript. BK, SD, and AM contributed to interpretation of the updated results and critically revised the manuscript. All authors read and approved the final manuscript.

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Data availability

All relevant data are included in the manuscript.

Declarations

Ethics approval and consent to participate

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Competing interests

The authors declare no competing interests.

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